E13 Multiple reactions in a non-isothermal batch reactor

The following reactions are taking place in the liquid phase in a 2000-L batch reactor:

A + 2B --> C	ΔH_{rxn}=-5 kcal/mol B	$-r_{1A} = k_{1A}*C_A*C_B^2$
3C + 2A --> D	ΔH_{rxn}=+10 kcal/mol C	$-r_{2A} = k_{2A}*C_A*C_C$
B + 3C --> E	ΔH_{rxn}=-50 kcal/mol B	$-r_{3C} = k_{3C}*C_B*C_C$

The initial temperature is 450 K and the initial concentrations of A, B, and C are 1.0, 0.5, and 0.2 M, respectively. The coolant temperature is constant at 400 K, and UA = 0.1 kcal/s*K. Given the following information, plot Ci and T as a function of time for up to 1 hr.

Data:

k1A = 1e-3 $M^{-2}*s^{-1}$
k2A = (1/3)e-3 $M^{-1}*s^{-1}$
k3C = 0.6e-3 $M^{-1}*s^{-1}$
All rate constant values given at 450 K
CPA = 0.01 kcal/mol/K
CPB = 0.01 kcal/mol/K
CPC = 0.05 kcal/mol/K
CPD = 0.08 kcal/mol/K
CPE = 0.05 kcal/mol/K

Decision Guide: Multiple Reactions in a Batch Reactor

Goal: Determine:

1. The concentration of each species as a function of time.
2. The reactor temperature as a function of time.
3. The behavior of a batch reactor containing multiple simultaneous reactions.

Step 1: Identify the Type of Problem

Ask yourself:

What reactor?
→ Batch Reactor

How many reactions?
→ Three reactions

Is temperature changing?
→ Yes

Is heat transfer present?
→ Yes

Because the reactor is: Non-isothermal and contains multiple reactions, you will need:

- One mole balance for each species.
- One energy balance.

Step 2: List All Species

Identify every species appearing in the reaction network: A, B, C, D, E

Each species may require its own mole balance and concentration equation.

Step 3: Write the Individual Reaction Rates

The problem provides:

$$r_1$$
$$r_2$$
r_3

for the three reactions. Do **not** combine these reactions. Treat each elementary reaction separately.

Step 4: Determine Which Reactions Create and Consume Each Species

Construct a species accounting table. Example thought process:

A:
Consumed in Reaction 1
Consumed in Reaction 2

B:
Consumed in Reaction 1
Consumed in Reaction 3

C:
Produced in Reaction 1
Consumed in Reaction 2
Consumed in Reaction 3

D:
Produced in Reaction 2

E:
Produced in Reaction 3

This step determines the form of each mole balance.

Step 5: Write a Mole Balance for Each Species

For a batch reactor use:

$$\frac{dC_{\mathrm{j}}}{dt} = r_{\mathrm{j}}$$

or equivalently begin with the batch-reactor mole balance:

$$\frac{dN_{\mathrm{j}}}{dt} = r_{\mathrm{j}}V$$

CBE 3610 Reactor Design Handbook Equation (14)

Write:

One ODE for A

One ODE for B

One ODE for C

One ODE for D

One ODE for E

using the information from Step 4.

Step 6: Determine the Heat Generated by Each Reaction

Each reaction has its own heat of reaction:
Reaction 1: Exothermic

Reaction 2: Endothermic

Reaction 3: Strongly Exothermic

Step 7: Build the Heat-Generation Term

The total heat generated in the reactor is the **sum of the heat contributions from all reactions**. For this example:

Reaction 1:
A + 2B → C
ΔHrxn,1 = -5 kcal/mol B

Reaction 2:
3C + 2A → D
ΔHrxn,2 = +10 kcal/mol C

Reaction 3:
B + 3C → E
ΔHrxn,3 = -50 kcal/mol B

Each reaction generates or consumes heat at a rate proportional to:
Reaction Rate × Heat of Reaction

Therefore: $\text{Total Heat Generated} = r_1 \Delta H_{\text{rxn,1}} + r_2 \Delta H_{\text{rxn,2}} + r_3 \Delta H_{\text{rxn,3}}$

More generally,

$$\text{Total Heat Generated} = \sum_{\text{i}} r_{\text{i}} \Delta H_{\text{rxn,i}}$$

The energy balance must include **all reactions**, not just the desired reaction, because every reaction contributes to the reactor temperature.

Physical Interpretation: In this problem:

- Reaction 1 is **exothermic** ($\Delta H\_rxn, 1 < 0$)
- Reaction 2 is **endothermic** ($\Delta H\_rxn, 2 > 0$)
- Reaction 3 is **strongly exothermic** ($\Delta H\_rxn, 3 < 0$)

Therefore:

Reaction 1 adds heat
Reaction 2 removes heat
Reaction 3 adds a large amount of heat

The reactor temperature is determined by the **net effect** of all three reactions occurring simultaneously.

Yes. I think the original wording is a little too brief. For students, I'd frame it in terms of **energy leaving the reactor**.

Step 8: Build the Heat-Removal Term

The problem provides:

- A coolant temperature, $T_\mathrm{a} = 400$ K
- A heat-transfer coefficient term, UA

These quantities indicate that the reactor exchanges heat with a cooling system. The rate of heat removal depends on the temperature difference between the reactor contents and the coolant:

$$\dot{Q}_\mathrm{r} = UA(T - T_\mathrm{a})$$

When $T > T_\mathrm{a}$ heat flows from the reactor to the coolant.

When: $T < T\_a$ heat flows from the coolant into the reactor.

Because the coolant temperature is lower than the initial reactor temperature in this example, the cooling system initially removes heat from the reactor.

Physical Interpretation

Hot Reactor
↓
Heat Transfer

$\downarrow$
Coolant

The larger the temperature difference, $(T - T_\mathrm{a})$, the faster energy is removed from the reactor.

The larger the value of UA, the more effective the cooling system is at removing heat.

Why This Matters: The reactor temperature is determined by a competition between:

Heat Generated by Reaction versus Heat Removed by Cooling

If heat generation exceeds heat removal, the reactor temperature rises. If heat removal exceeds heat generation, the reactor temperature falls. The energy balance tracks this competition over time.

Step 9: Write the Energy Balance

Use: **CBE 3610 Reactor Design Handbook Equation (41)**

$$\frac{dT}{dt} = \frac{V \sum_i r_\mathrm{i} \Delta H_\mathrm{rxn,i} - UA(T - T_\mathrm{a})}{\sum_j N_\mathrm{j} C_\mathrm{Pj}}$$

which follows from the multiple-reaction batch-reactor energy balance. This becomes the temperature ODE.

Step 10: Count Equations and Unknowns

You should now have:

Unknowns: CA, CB, CC, CD, CE, T
Governing Equations: 5 Species Balances, 1 Energy Balance

The mathematical model is now complete.

Step 11: Apply Initial Conditions

CA(0), CB(0), CC(0), T(0): given in the problem statement.

For species not initially present:

CD(0)=0, CE(0)=0

Step 12: Solve Numerically
The resulting system contain coupled ODEs because:

- Reaction rates depend on concentrations.
- Heat generation depends on reaction rates.
- Temperature influences kinetics.
- Concentrations influence temperature.

A numerical solver is generally required.

Final Answer Checklist

☑ Identified all species.
☑ Wrote all reaction-rate expressions.
☑ Constructed one mole balance for each species.
☑ Included the contribution of all reactions in the energy balance.
☑ Included heat transfer to the coolant.
☑ Applied initial conditions.
☑ Solved the coupled ODE system.
☑ Generated plots of $C_{\mathrm{i}}(t)$ and $T(t)$.

CBE 3610 Reactor Design Handbook Equations Needed

Eq. (14) Batch Reactor Mole Balance

Eq. (41) Batch Reactor Energy Balance
(Multiple Reactions)

Key Insight

Reaction Rates
↓
Change Concentrations
↓
Change Heat Generation
↓
Change Temperature
↓
Change Reaction Rates

Unlike single-reaction systems, every reaction contributes to both the species balances and the energy balance. The resulting concentration and temperature profiles emerge from the interaction of the entire reaction network.